Part IV

Fermat



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Pierre Fermat (c. 1601–1665) has been called the greatest amateur mathematician. After growing up in Beaumont-de-Lomagne in Gascony (where his home now houses an interesting museum**), he studied in Orléans and Toulouse, became "commissioner of requests" in 1631, and *conseiller du roi* in the local *parlement*, through which any petitions to the king had to pass. He died in Castres, where he was in the commission implementing the *Édit de Nantes*, which gave some protection to the persecuted protestant Huguenots. Fermat never left the area, never published a paper, and still became the second-best mathematician of his century (after Newton). Fermat communicated his mathematical discoveries in numerous letters, usually without proof and often in the form of challenges, to his contemporaries. (Among them was René Descartes, who could only be reached through his friend Marin Mersenne in Paris, because for many years he lived in Holland without a fixed address—a Flying Dutchman of mathematics, like the modern-day late Pál Erdős.)



Fermat was a pioneer in several areas. His method for drawing a tangent to certain plane curves was a step in the invention of calculus—later came Newton and Leibniz. He invented probability theory, in extensive correspondence with Blaise Pascal around 1654. He determined extrema of functions as zeroes of their derivative, and used this to calculate the path of light through different media according to the "principle of least time". There was a controversy between him and Descartes about the discovery of analytical geometry; certainly Fermat was the first to use it in three dimensions.

But Fermat's greatest contributions—and those of interest for computer algebra—were in number theory. He was fascinated by perfect and amicable numbers, and the Pell–Fermat equation $x^2 - ny^2 = 1$. Fermat discovered that primes of the form $4n+1$ can be represented (in exactly one way) as a sum of two squares; for example, $29 = 5^2 + 2^2$. (It is easy to show that numbers of the form $4n-1$ are never sums of two squares; see Exercise 18.1.) His "method of infinite descent" can determine the (un)solvability of many Diophantine equations.

** worth the detour

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Fermat wrote to Bernard Frénicle de Bessy around August 1640 that the numbers $2^{2^0}+1=3$, $2^{2^1}+1=5$, $2^{2^2}+1=17$, $2^{2^3}+1=257$, and $2^{2^4}+1=65\,537$ are primes; he conjectured that all these **Fermat numbers** $F_n=2^{2^n}+1$ were prime. He was wrong; the next values of F_n, at least up to $n=23$, are not prime; see Section 4.3. Not much harm done; he pointed out that he did not have a proof of his conjecture. We come across Fermat numbers in Chapter 8, where they are used in the integer Fourier Transform, and Chapter 18. Weil (1984) presents Fermat's achievements in detail.



The theorem that includes Fermat into our little Hall of Fame is that for a prime p and an integer a, $a^{p-1}-1$ is divisible by p. He stated it in a letter to Frénicle on 18 October 1640: *Tout nombre premier mesure infailliblement une des puissance* -1 *de quelque progression que ce soit, et l'exposant de la dite puissance est sous-multiple du nombre premier donné* -1.[1] He forgot to mention that we have to disallow the geometric progression $a, a^2, a^3, \ldots$ if p divides a. We will call it and various generalizations "Fermat's little theorem" in this book; they play a crucial role in primality testing and the factorization of polynomials and integers (Chapters 14, 18, and 19). Leibniz rediscovered and proved this result in unpublished notes from 1680; see Notes 4.4. Euler (1732/33, 1747/48) was the first to publish a proof. He also derived conditions on the factors of Fermat numbers, which led him to the factor 641 of F_5.

Finally, there is Fermat's (in)famous remark in the margin near the eighth problem of Book II of Bachet's translation of Diophantus' *Arithmetic*, which deals with rational solutions of the equation $x^2+y^2=z^2$: *Cubum autem in duos cubos, aut quadratoquadratum in duos quadratoquadratos & generaliter nullam in infinitum ultra quadratum potestatem in duos eiusdem nominis fas est dividere; cuius rei demonstrationem mirabilem sane detexi. Hanc marginis exiguitas non*

[1] Every prime number divides invariably one of the powers -1 in any given geometric progression, and the exponent of this power is a divisor of the given prime number -1.

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caperet.[2] A proof of this eluded mathematicians for over three centuries and inspired Kummer's theory of ideals and the construction of large parts of the edifice of *arithmetic algebraic geometry*, culminating in Wiles' proof of "Fermat's last theorem" via a special case of the Taniyama–Weil conjecture (Wiles 1995, Taylor & Wiles 1995; see van der Poorten 1996 for the mathematics and Singh 1997 for the story). The designation comes from the fact that after all of Fermat's claims had been proven, this remained open as the last one.

His son, Samuel de Fermat, published Diophant's *Arithmetic* with Fermat's annotations, and in 1679 the *Varia opera mathematica D. Petri de Fermat, Senatoris Tolosani.* His dedication to Ferdinand II. von Fürstenberg, featured on page 513, reads: *To His Highness Prince Ferdinand, Bishop of Paderborn, Coadiutor of Münster, Duke of Pyrmont, Free Baron of Fürstenberg. By Samuel de Fermat S. P. Motto: soft and strong*. Ferdinand II. (1626–1683) was a shining light of Paderborn science, student at the University of Paderborn from 1644 to 1646, erudite author of *Monumenta Paderbornensia* on the local history, correspondent with the leading philosophers and scientists of his times, and sponsor of the arts and architecture. His residence is shown in Figure 13.4. Fermat's dedication makes it plausible that he financed the *Varia opera.*

Samuel de Fermat includes a poem, whose title and first lines are:

> *De Principis eiusdem præclaro*
> *Monumentorum Paderbornensium opere.*
>
> *Dum Paderæ fontes æterno carmine Princeps*
> *Aonij celebrat spes columenque chori,*
> *Ut superat quæ sic ponit monumenta, suisque*
> *Altius ipse aliud tollit ad astra modis!*[3]

[2] But it is impossible to divide a cube into two cubes, or a fourth power into two fourth powers, or generally any power beyond the squares into two like powers; I discovered a truly marvelous proof of this fact. The margin is too narrow to write it down.

[3] On the Prince's famous work *Monumenta Paderbornensia*: The Prince, hope and pillar of the chorus of the Muses, celebrates the sources of the Pader river with his eternal song. Just as he who builds such monuments towers high, he carries in his way [through his generosity] another work [Fermat's *Opera*] higher up to the stars!

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Il est remarquable qu'on déduise ainsi du calcul intégral
une propriété essentielle des nombres premiers; mais
toutes les vérités mathématiques sont liées les unes aux autres,
et tous les moyens de les découvrir sont également admissibles.[1]

Adrien-Marie Legendre (1830)

Can I get a witness

The Rolling Stones (1964)

No mathematician can now write on demand a prime with,
say, 10 million digits, although one surely exists.

Stanisław Marcin Ulam (1964)

Such sentiments [half-credence in the supernatural] are seldom
thoroughly stifled unless by reference to the doctrine of chance, or, as
it is technically termed, the Calculus of Probabilities. Now this
Calculus is, in its essence, purely mathematical; and thus we have the
anomaly of the most rigidly exact in science applied to the shadow and
spirituality of the most intangible in speculation.

Edgar Allan Poe (1842)

"Du har sagt det där förut", sa Kollberg torrt.
"Det är rena gissningen."—"Sannolikhetsprincipen."[2]

Maj Sjöwall and Per Wahlöö (1967)

[1] It is remarkable that one should deduce in this way from integral calculus an essential property of the prime numbers; but all mathematical truths are connected to each other, and all means of discovering them are equally admissible.

[2] "You have said that before," Kollberg said drily. "It is pure guesswork."—"The principle of probability."

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